

DOE ALCC

*ASCR Leadership Computing Challenge — Frontier, Aurora, Polaris, NERSC allocations
aligned to DOE mission*

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Program overview

The *ASCR Leadership Computing Challenge* (ALCC) is the second of the two large open allocation programs run by the U.S. Department of Energy Office of Science. It is administered by the Office of *Advanced Scientific Computing Research* (ASCR) and is the mission-aligned counterpart to INCITE. Where INCITE is curiosity-driven and open to any domain that can defend a leadership-scale science case, ALCC is explicitly steered toward the DOE mission portfolio. Energy production and storage, climate and Earth-system modeling, nuclear security, fusion and plasma science, accelerator design, and scientific artificial intelligence are the recurring priority themes.

ALCC awards approximately *30 percent* of the available compute time on the DOE leadership-class systems. The eligible machines are the OLCF Frontier system at Oak Ridge, the ALCF Aurora and Polaris systems at Argonne, and the NERSC Perlmutter system at Lawrence Berkeley. Awards are one-year allocations that begin on *July 1* and run through *June 30* of the following year. The program complements INCITE by providing a second, mission-targeted entry point on the same hardware in the same calendar.

Hardware specs

The four ALCC-eligible systems span three accelerator generations. Frontier is *AMD MI250X*, Aurora is *Intel Ponte Vecchio* (Data Center GPU Max 1550), and the two NERSC and ALCF Ampere systems use *NVIDIA A100* in slightly different memory configurations.

System	Accelerator	VRAM	FP16/ BF16 (TFLOPS, dense)	Mem BW (TB/s)	Interconnect	Nodes (ALCC-eligible)
Frontier (OLCF)	AMD MI250X (per GCD)	128 GB	383	3.2	Sling-shot-11, 4 NICs/node	9{,}408
Aurora (ALCF)	Intel PVC Max 1550	128 GB	839	3.3	Sling-shot-11, 8 NICs/node	10{,}624

System	Accelerator	VRAM	FP16/BF16 (TFLOPS, dense)	Mem BW (TB/s)	Interconnect	Nodes (ALCC-eligible)
Polaris (ALCF)	NVIDIA A100 40 GB	40 GB	312	1.56	Sling-shot-10, HPE Apollo	560
Perlmutter GPU (NERSC)	NVIDIA A100 40/80 GB	40 / 80 GB	312	1.56 / 2.0	Sling-shot-11, 4 NICs/node	1{,}792

Frontier and Aurora are *exascale* machines and dominate the available aggregate FLOPs in the ALCC pool; Polaris and Perlmutter remain attractive for projects that need NVIDIA-CUDA tooling at production scale.

Allocation sizes and H200-hour equivalents

Typical ALCC awards range from roughly $50{,}000$ node-hours for a focused single-system project to $500{,}000$ node-hours for a flagship mission-aligned campaign. Conversion to H200-hour equivalents uses the dense FP16/BF16 tensor throughput ratio against the H200 reference value of 989 TFLOPS, multiplied by the GPUs per node on the awarded system.

System	GPUs/node	Per-GPU FP16 TFLOPS	Node-hours	Per-node ratio vs H200	H200-hour equivalent
Frontier (small, 50k nh)	8 GCDs	383	$50{,}000$	$8 \times \frac{383}{989} = 3.10$	$\approx 154{,}900$
Frontier (large, 500k nh)	8 GCDs	383	$500{,}000$	3.10	$\approx 1{,}549{,}000$
Aurora (small, 50k nh)	6 tiles	839	$50{,}000$	$6 \times \frac{839}{989} = 5.09$	$\approx 254{,}500$
Aurora (large, 500k nh)	6 tiles	839	$500{,}000$	5.09	$\approx 2{,}544{,}500$
Polaris (small, 50k nh)	4 A100	312	$50{,}000$	$4 \times \frac{312}{989} = 1.26$	$\approx 63{,}100$
Polaris (large, 500k nh)	4 A100	312	$500{,}000$	1.26	$\approx 631{,}000$
Perlmutter (small, 50k nh)	4 A100	312	$50{,}000$	1.26	$\approx 63{,}100$
Perlmutter (large, 500k nh)	4 A100	312	$500{,}000$	1.26	$\approx 631{,}000$

The headline takeaway is that an ALCC award sits in the same *millions of H200-hours* tier as INCITE. A 500{,}000 node-hour Aurora allocation alone exceeds 2.5 million H200-hours of equivalent FP16 throughput.

Approval timeline

ALCC runs on a strict annual cycle synchronized with the DOE fiscal calendar. The call for proposals opens in *February*, with a submission deadline in *early April*. ASCR convenes a peer-review panel through April and May. Awards are announced in *June*, and the allocation begins on *July 1* and runs through *June 30* of the following year.

The proposal package itself is a *10-page* science and computational narrative, accompanied by a separate *computational readiness* document that demonstrates the code can use the targeted system at scale, and a *data management plan* describing storage, retention, and release. ASCR reviews on scientific merit, mission relevance, and technical readiness; the readiness component routinely eliminates otherwise strong science proposals when the team cannot show prior scaling runs.

Eligibility

ALCC is open to researchers at *U.S. national laboratories, universities, industry, and other federal agencies*. Foreign collaborators are permitted on the project team, although the principal investigator and the lead institution typically must be U.S.-based. The PI is usually a senior researcher with a track record of leadership-scale runs, and graduate students and postdocs participate as co-investigators or as named team members. The work itself must align with a stated DOE mission area; method-development work without a clear mission anchor is generally re-directed to INCITE.

Things to know

ALCC has a strong revealed preference for *applied, mission-relevant science* over basic method development. Reviewers want to see how the compute moves a DOE program forward this fiscal year, not how it advances an algorithm in the abstract. Because ALCC and INCITE compete for the same physical machines, queue dynamics in the second half of an award year can be tight, and projects that cannot absorb that risk should plan for it explicitly.

A practical pre-application path is to first acquire *Director's Discretionary* (DD) hours at the target facility, use them to run the scaling and readiness studies, and cite those runs in the ALCC proposal. *Scientific AI*, particularly large-model training and inference for drug discovery, materials, climate emulation, and fusion control, is an increasingly competitive sub-portfolio inside ALCC and now routinely captures awards that would historically have gone to traditional simulation.

Strategic fit

For a single graduate student without an established DOE-aligned advisor, ALCC is *not a realistic first application*. It sits at the same tier as INCITE in terms of scale and review depth, and the mission-alignment requirement adds an additional constraint that is hard to satisfy from outside an ongoing DOE program. The realistic path is identical to INCITE. The student joins an advisor-led DOE-funded thread, contributes to the readiness runs on Director’s Discretionary time, and is named on the ALCC proposal as a co-investigator. The reward is also identical, namely access to *millions of H200-equivalent hours* on a leadership-class system for a full year.

ALCC vs INCITE vs DD

Program	Tier	Share of machine	Approval difficulty	Typical award	One-line summary
INCITE	Leadership	$\approx 60\%$	Very high	$10^5\text{--}10^6$ node-hours	Open, curiosity-driven, peer-reviewed at the highest bar.
ALCC	Leadership	$\approx 30\%$	High	$5 \times 10^4\text{--}5 \times 10^5$ node-hours	Mission-aligned counterpart to INCITE on the same machines.
Director’s Discretionary	Per-facility	$\approx 10\%$	Low to moderate	$10^3\text{--}10^4$ node-hours	Facility-director’s pool for startup, scaling, and readiness work.

Summary

System	Typical ALCC allocation	H200-hour equivalent (range)	Fit rating
Frontier (OLCF)	50k–500k node-hours	$\approx 1.5 \times 10^5 - 1.5 \times 10^6$	Stretch with DOE-aligned advisor
Aurora (ALCF)	50k–500k node-hours	$\approx 2.5 \times 10^5 - 2.5 \times 10^6$	Stretch with DOE-aligned advisor

System	Typical ALCC allocation	H200-hour equivalent (range)	Fit rating
Polaris (ALCF)	50k–500k node-hours	$\approx 6.3 \times 10^4 - 6.3 \times 10^5$	Plausible via co-investigator role
Perlmutter GPU (NERSC)	50k–500k node-hours	$\approx 6.3 \times 10^4 - 6.3 \times 10^5$	Plausible via co-investigator role

H200-hour equivalence is computed at dense FP16/BF16 tensor throughput (989 TFLOPS, 141 GB HBM3e, 4.8 TB/s) per H200 SXM GPU. Conversions below use the per-hour FP16 TFLOP ratio; memory-bound workloads scale differently and the bandwidth column is provided so readers can re-derive their own equivalence.

Sources

- DOE Office of Science, ASCR Leadership Computing Challenge program page, science.osti.gov/ascr/alcc (verified 2024–2026).
- Oak Ridge Leadership Computing Facility, Frontier system specifications, olcf.ornl.gov/frontier (verified 2024–2026).
- Argonne Leadership Computing Facility, Aurora and Polaris system documentation, alcf.anl.gov (verified 2024–2026).
- National Energy Research Scientific Computing Center, Perlmutter architecture documentation, docs.nersc.gov/systems/perlmutter (verified 2024–2026).
- ALCC FY2024 and FY2025 awards announcements, science.osti.gov/ascr (verified 2024–2026 via WebSearch and WebFetch).
- NVIDIA H200 SXM datasheet, used as the H200-hour reference (989 TFLOPS dense FP16/BF16 tensor, 141 GB HBM3e, 4.8 TB/s).